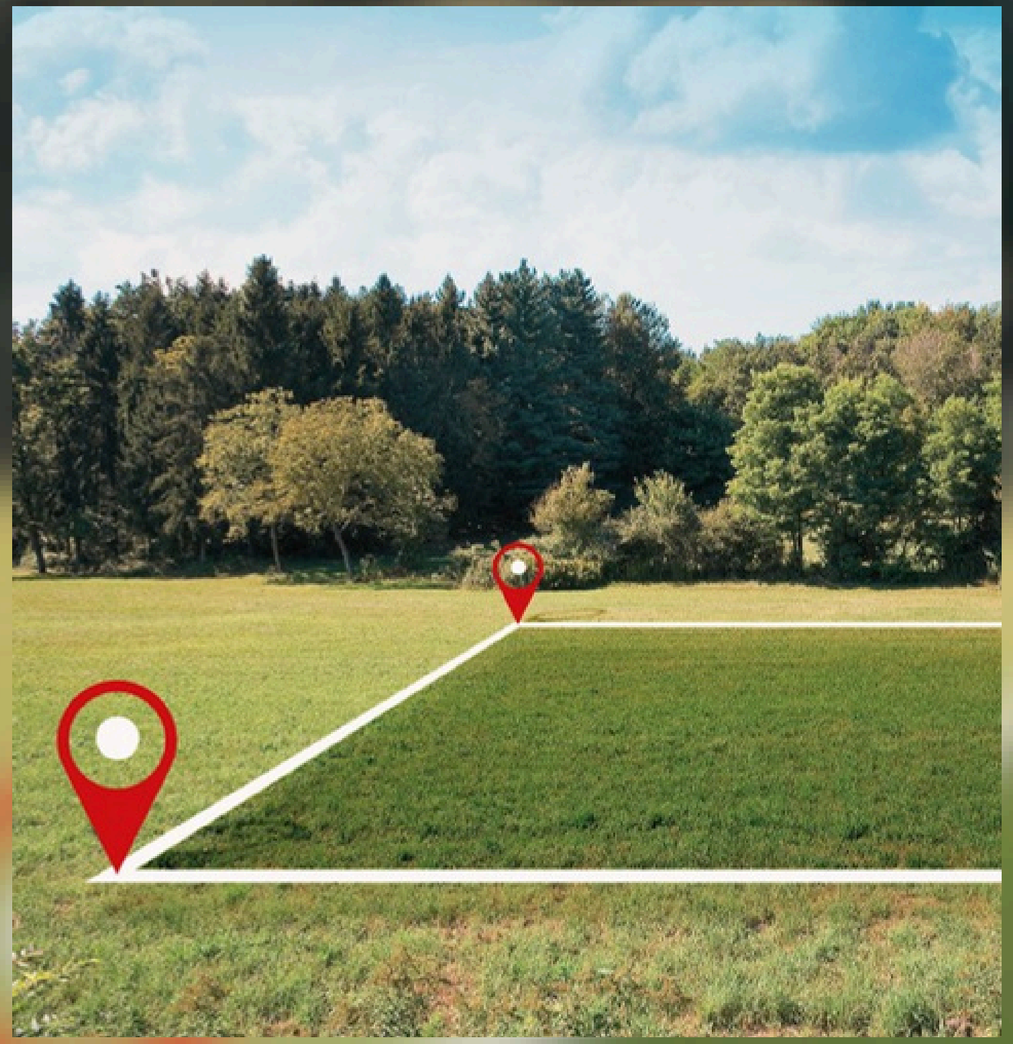
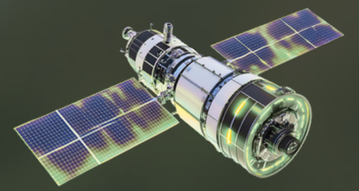


Manouba School of Engineering



PCA–DWT Based Land Classification of Kairouan from Sentinel-2 Data

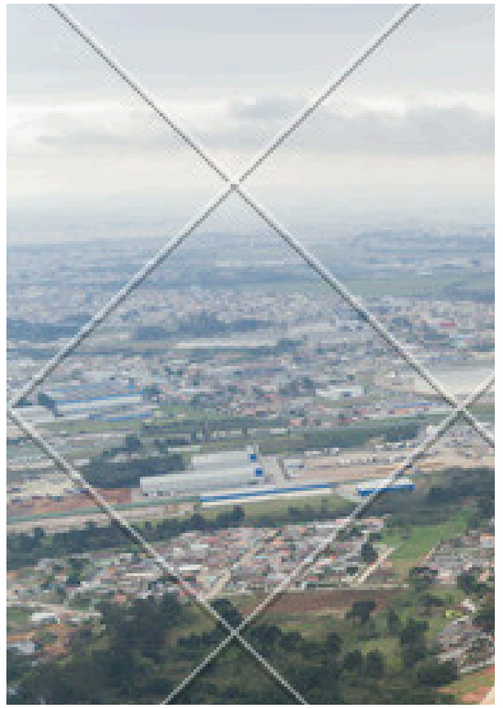
11 December, 2025

Project Overview

- Introduction
- STEP 1: Feature Extraction and Statistics
- STEP 2: Classification and Accuracy Assessment
- Conclusion

INTRODUCTION

- Objective: Classify land cover types in the Kairouan region using Sentinel-2 TCI imagery (June 2019).
- Approach: Apply Principal Component Analysis (PCA) and Discrete Wavelet Transform (DWT) for feature extraction.
- Goal: Improve land classification accuracy through advanced spectral and spatial transformations.



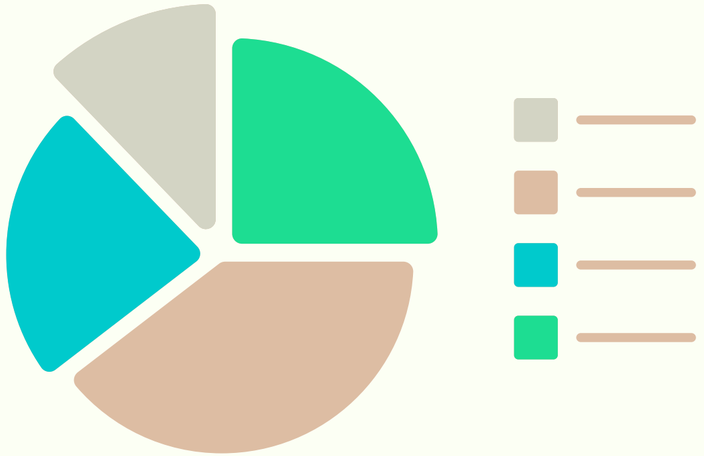




STEP 1

Feature Extraction and Statistics

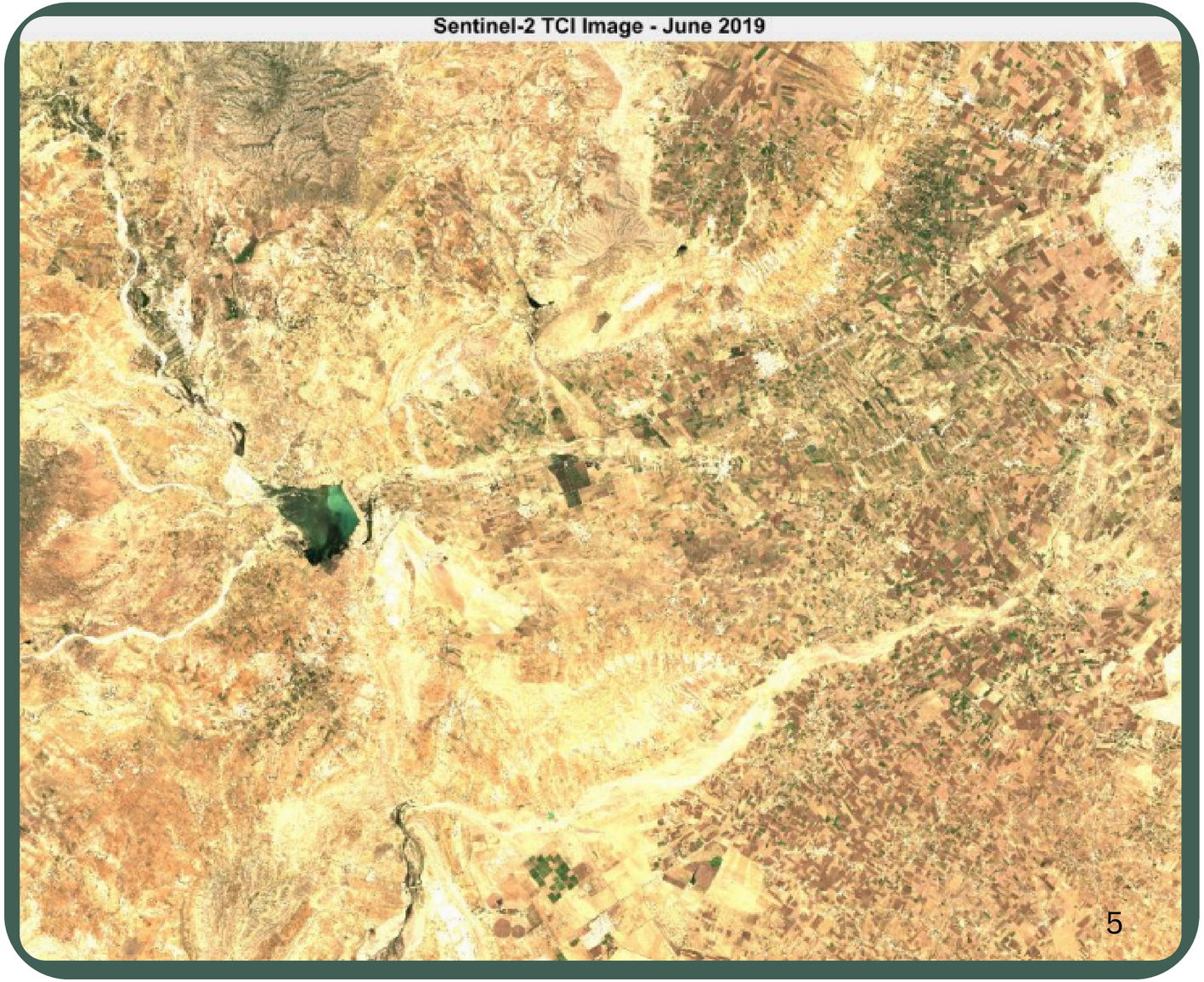




STEP 1.1

Image loading

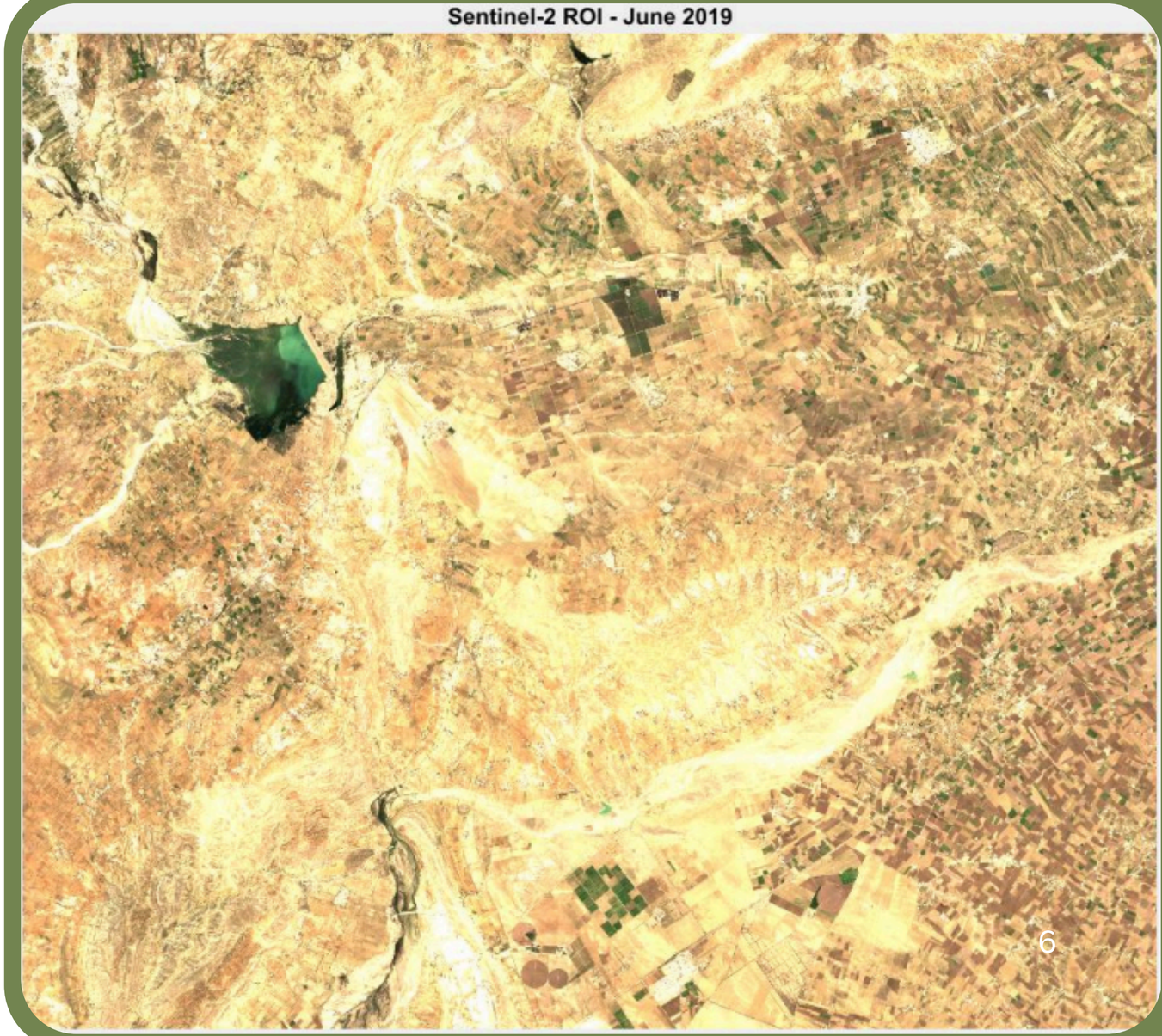
Loads the Sentinel-2 RGB image into MATLAB and displays it for initial inspection.



STEP 1.2

Region of Interest Extraction

A 3000×3000 pixel Region of Interest is cropped from the full image.



1

Convert and reshape the data

Convert the image to double precision and reshape it into a 2D matrix (pixels × bands).

2

Center the data

Subtract the mean of each band from all pixels.

3

Compute the covariance matrix

Calculate how the spectral bands vary together.

.

4

Compute eigenvalues and eigenvectors

Perform eigen-decomposition of the covariance matrix.

5

Sort components by importance

Sort eigenvalues from highest to lowest.

6

Project data onto the new components

Transform the centered data using the eigenvectors.

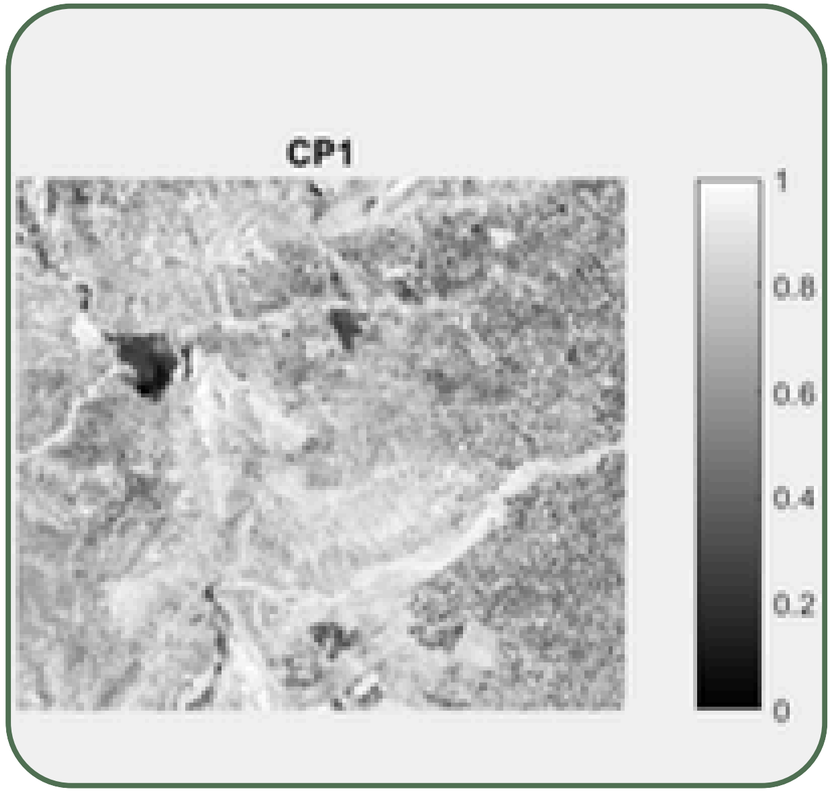
STEP 1.4

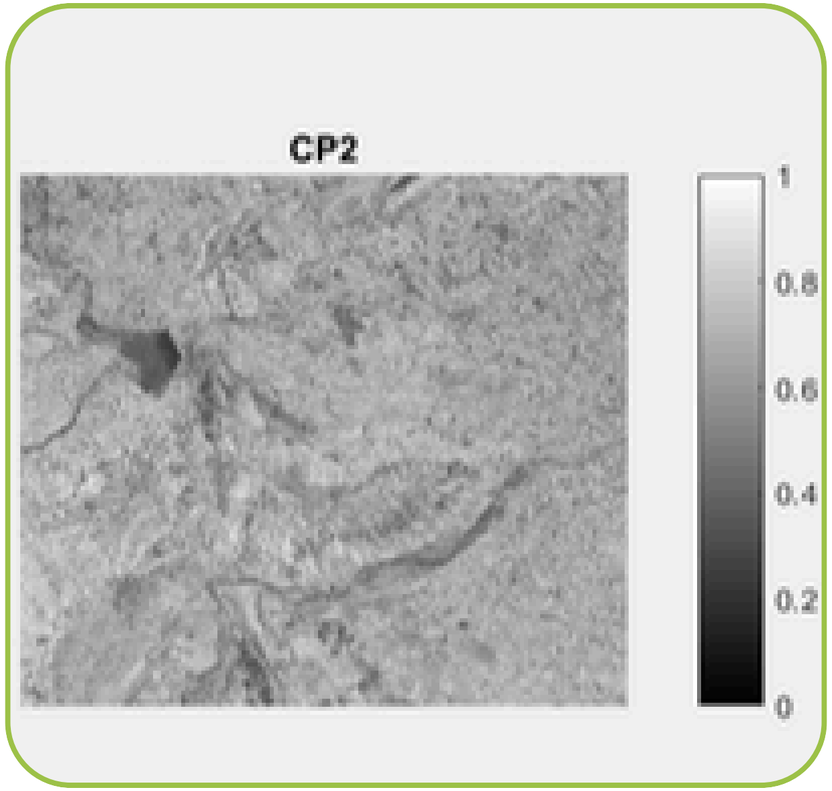
Energy Calculation

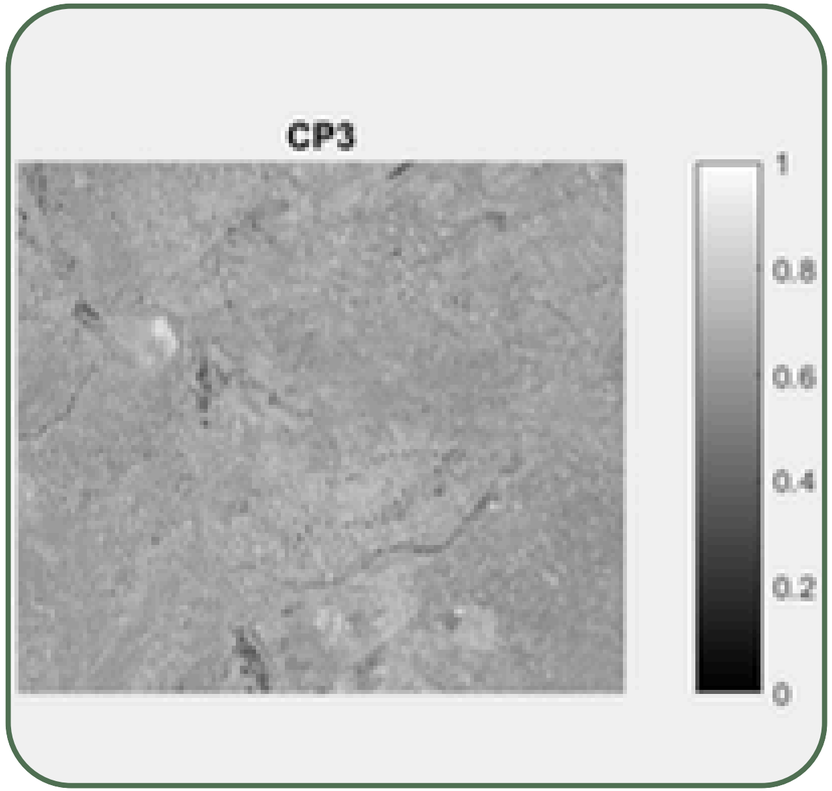
	CP1	CP2	CP3
Eigenvalue	4629.63	389.06	81.46
Individual Energy	90.74%	7.66%	1.60%
Cumulative Energy	90.74%	98.40%	100%
Interpretation	Captures the dominant landscape	Captures secondary variations	Contains mostly noise or minor variations

STEP 1.4

PCA Results







STEP 1.5

DISCRETE WAVELET TRANSFORM

01

sym2

Symmetrical
Good for general features

VS

02

db45

More complex
Captures finer details

STEP 1.5

DWT Components

01 Approximation (A)

Low-frequency content (general shapes and brightness)

02 Horizontal (DH)

Horizontal edge details.

03 Vertical (DV)

Vertical edge details.

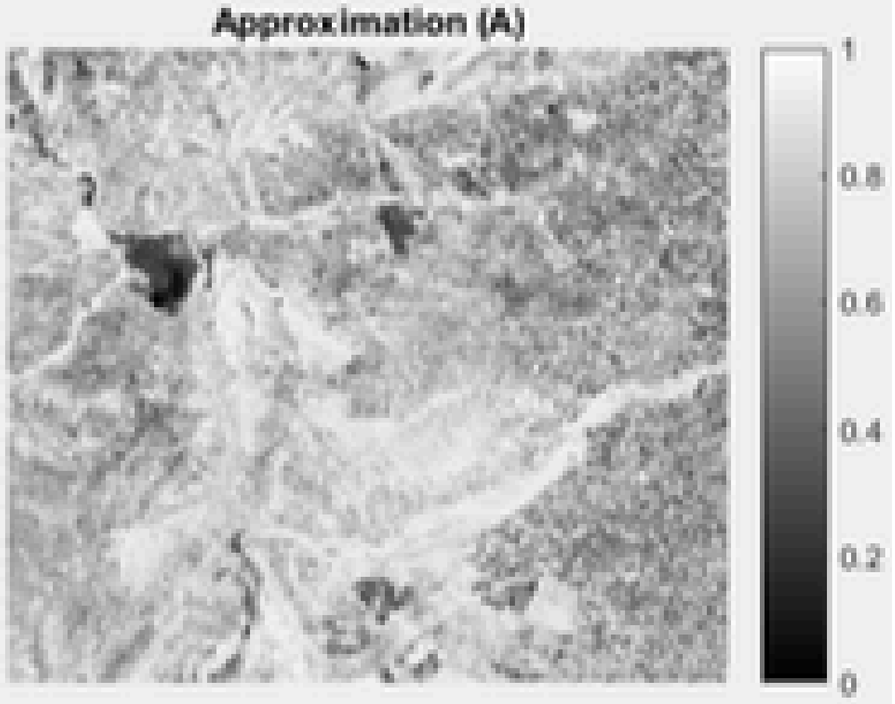
04 Diagonal (DD)

Diagonal edge details

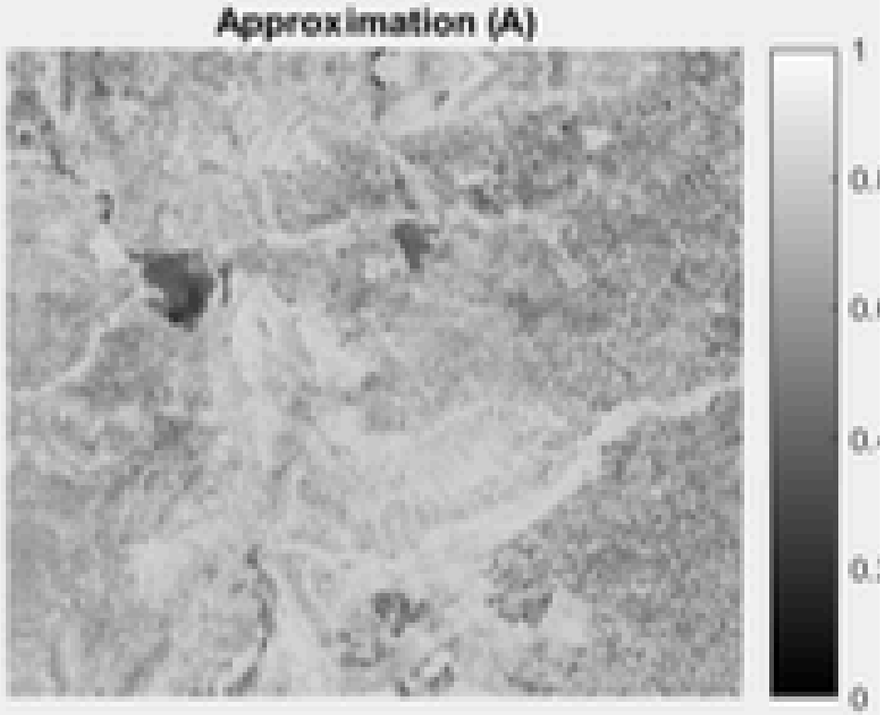
STEP 1.5

DWT Results

sym2

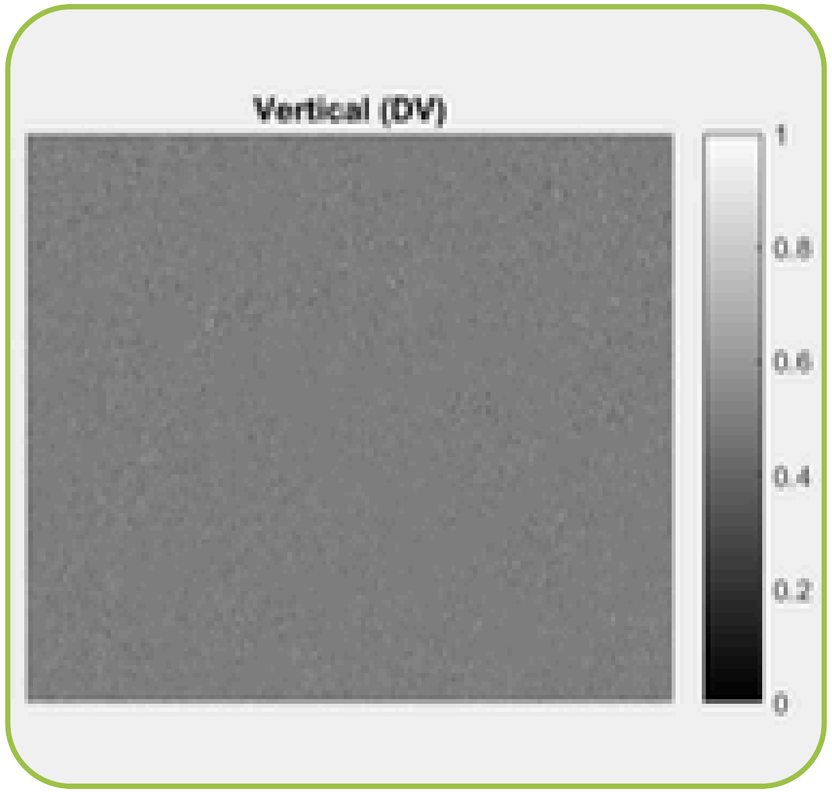


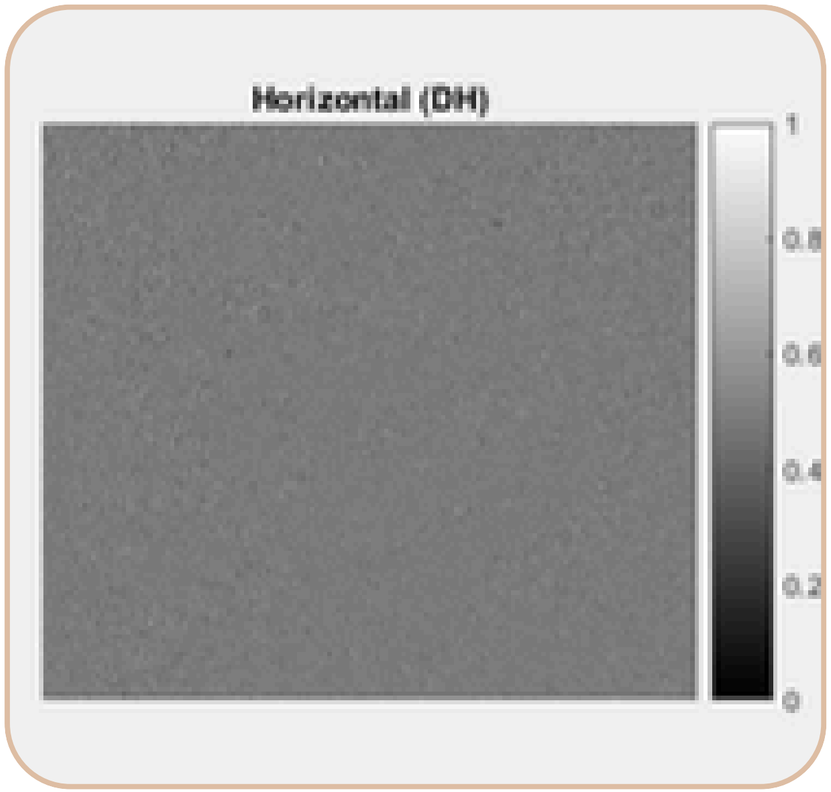
db45

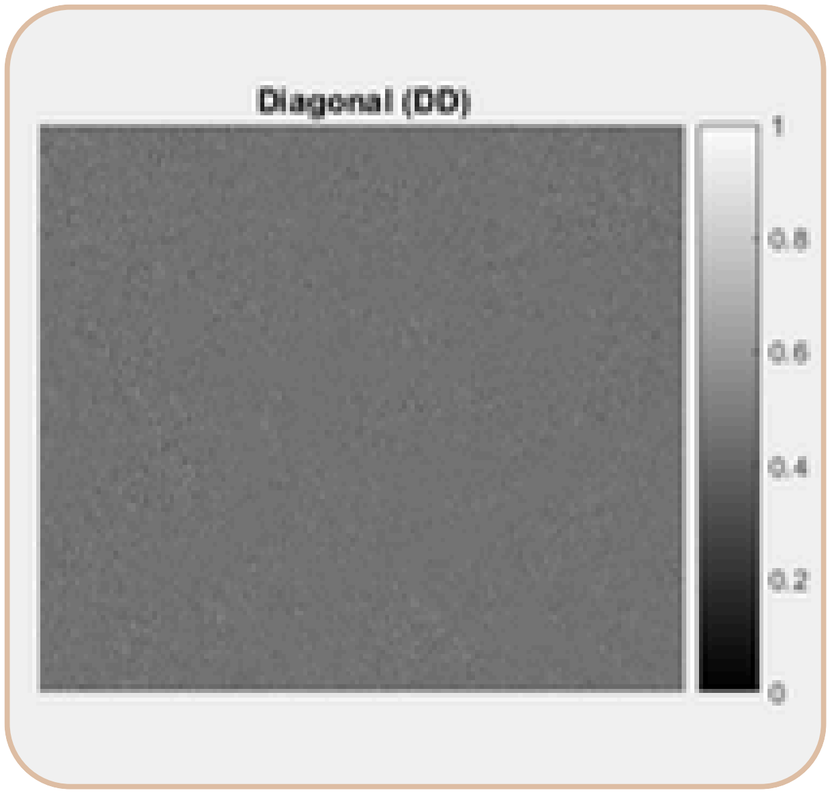


STEP 1.5

DWT sym2 Results

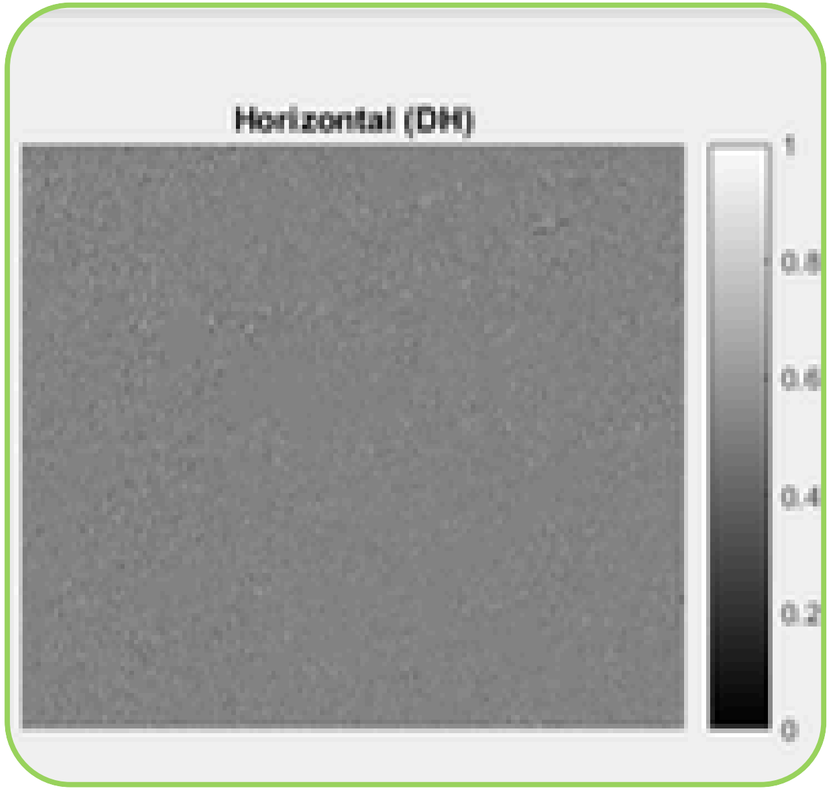


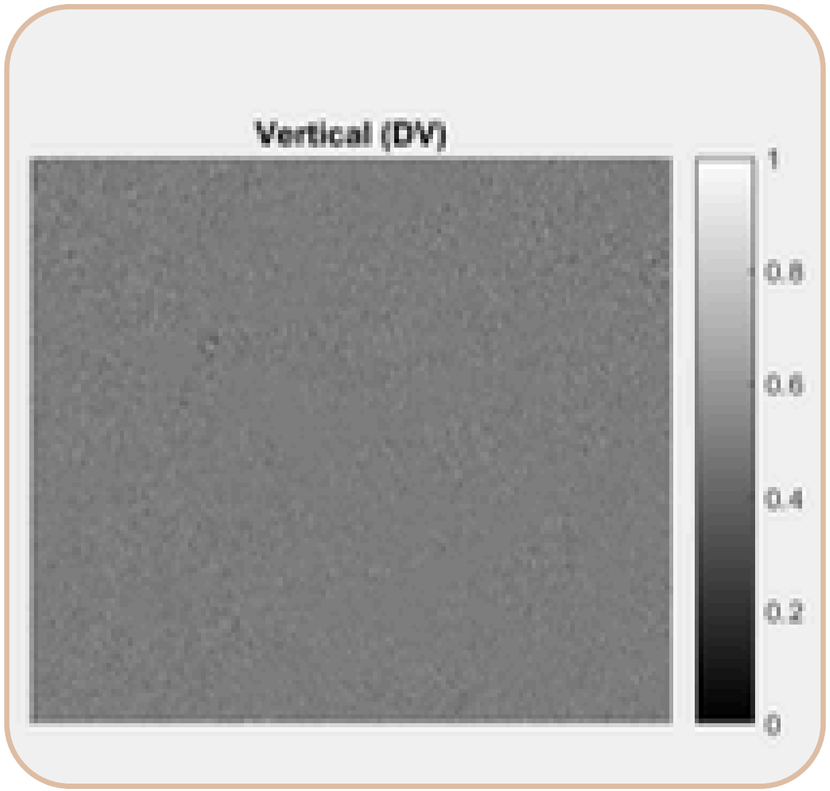


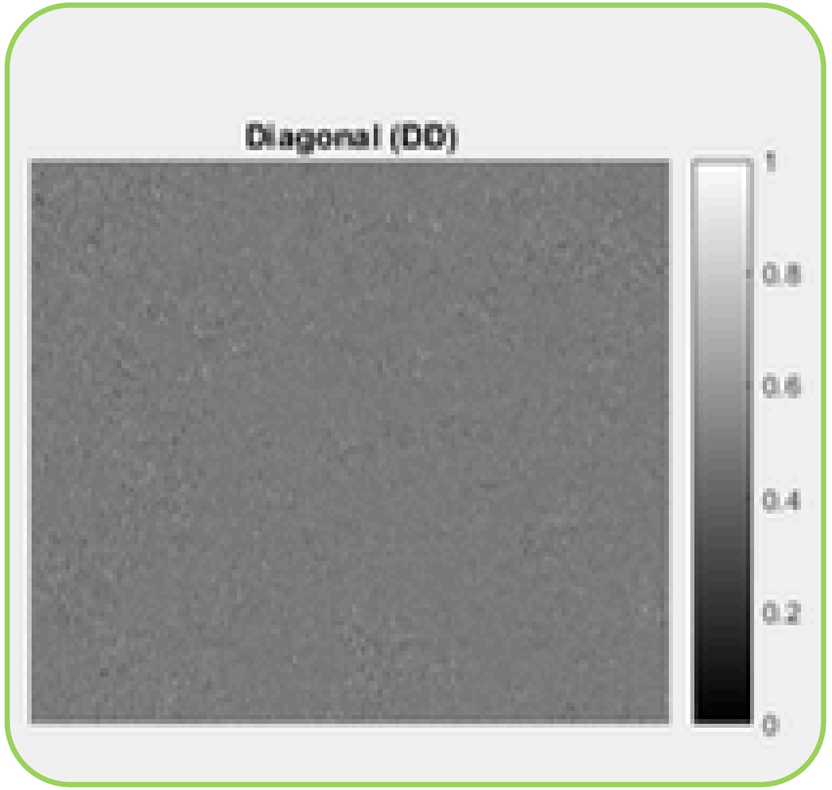


STEP 1.5

DWT db45 Results

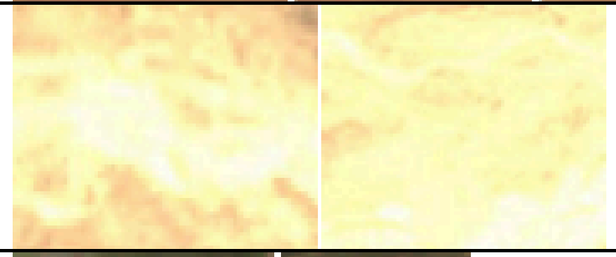




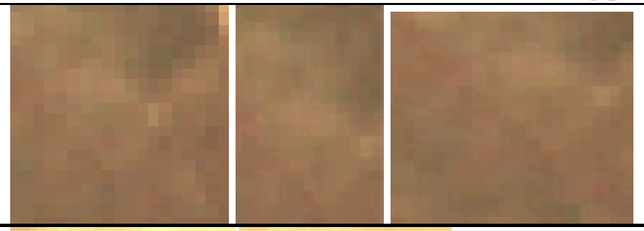
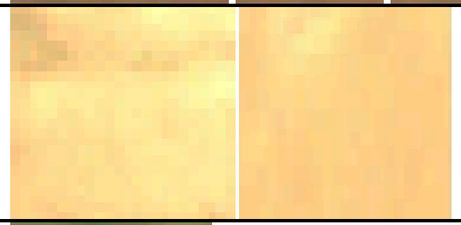
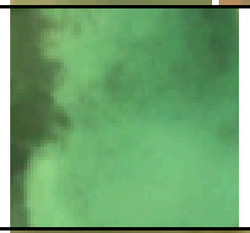


STEP 1.6

Definition Of The 10 Coordinates Table

Classes	Textures	Exemples de fenêtres extraites de l'image TCI – Sentinel-2 de la date du 05.06.2019
1	Urbain	
2	Sol nu	
3	Sol nu couvert de pailles	
4	Oued	
5	Oliviers	

STEP 1.6 Definition Of The 10 Coordinates Table

Classes	Textures	Exemples de fenêtres extraites de l'image TCI – Sentinel-2 de la date du 05.06.2019
6	Jachère	
7	Céréales	
8	Maraîchère	
9	Barrage	
10	Arboriculture	

STEP 1.6

Examples of Manual Selection





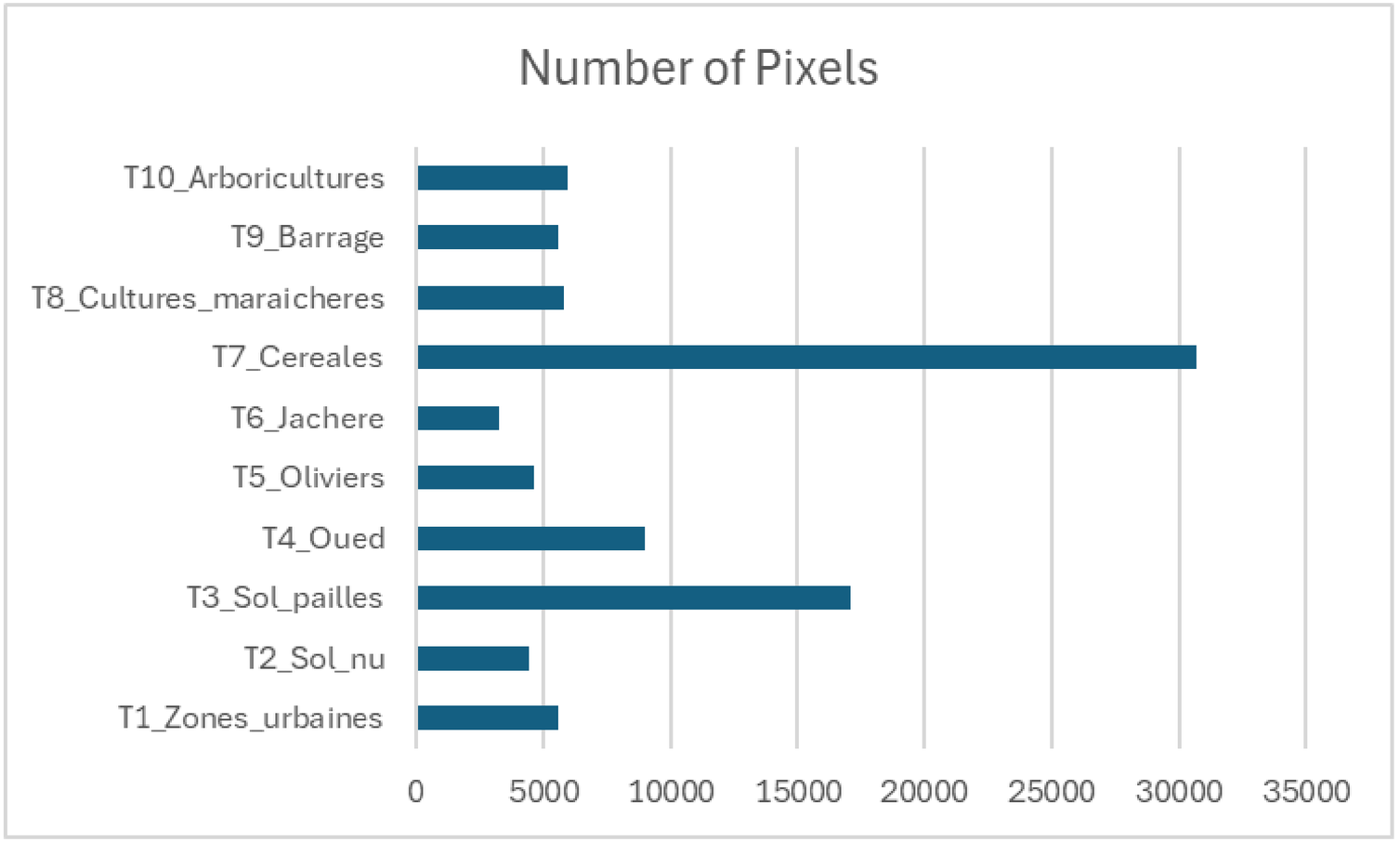
STEP 1.6

Color Code Table

Texture	Label	Couleur	Rouge	Vert	Bleu
Zones urbaines	T1		255	255	255
Sol nu	T2		000	000	000
Sol nu couvert de pailles	T3		255	255	000
Oued	T4		166	230	204
Oliviers	T5		000	255	000
Jachère	T6		166	000	204
Céréales	T7		230	230	000
Cultures maraîchères	T8		255	000	000
Barrage	T9		000	000	255
Arboricultures	T10		230	128	000

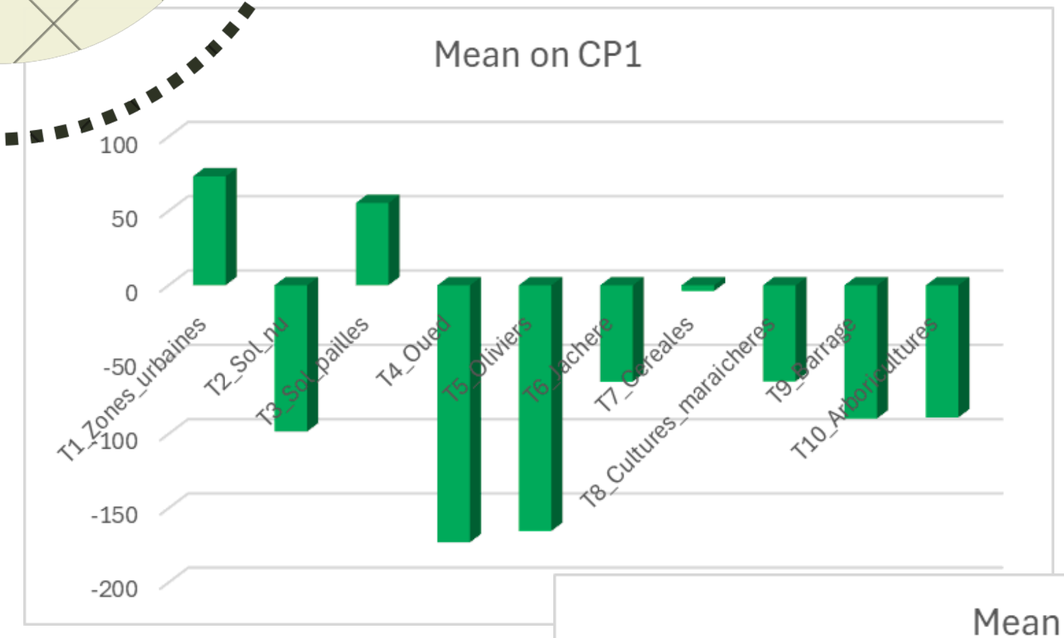
STEP 1.6

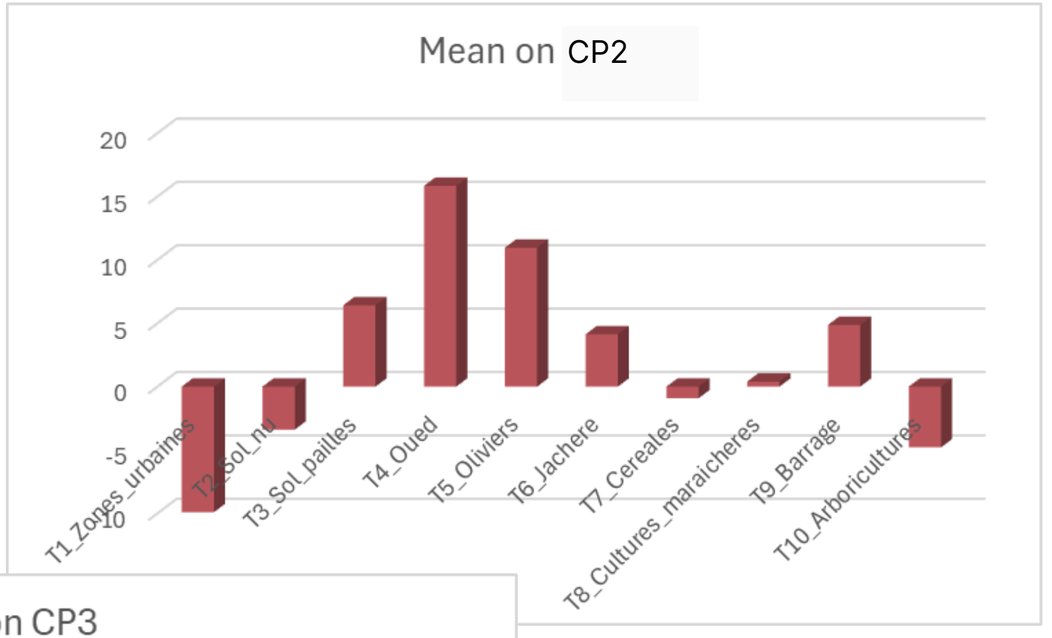
Statistics

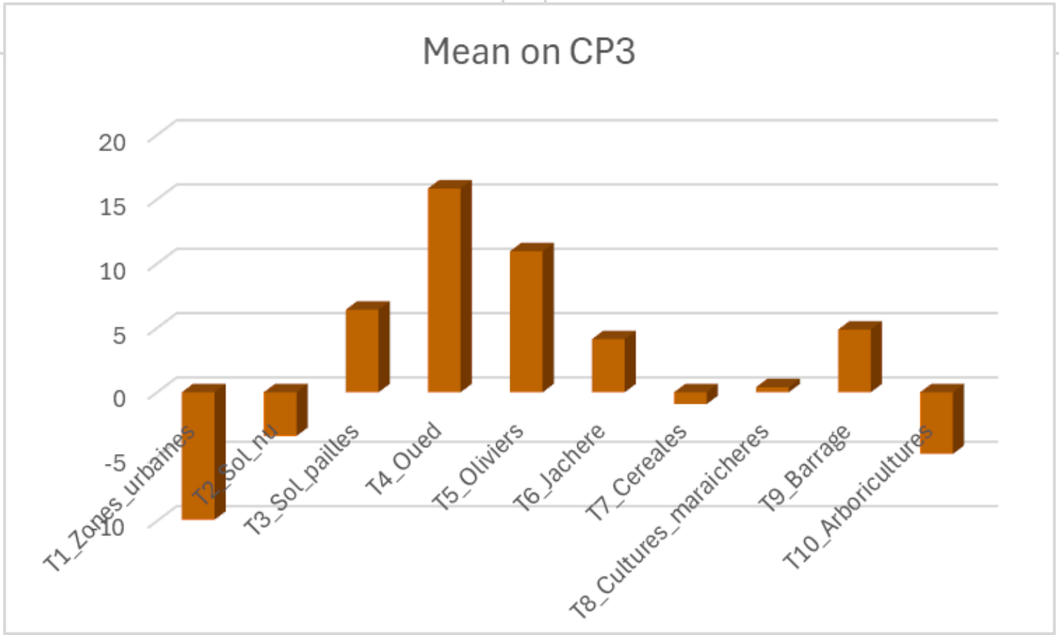


STEP 1.6

Statistical Signatures – PCA Components (CP1, CP2, CP3)

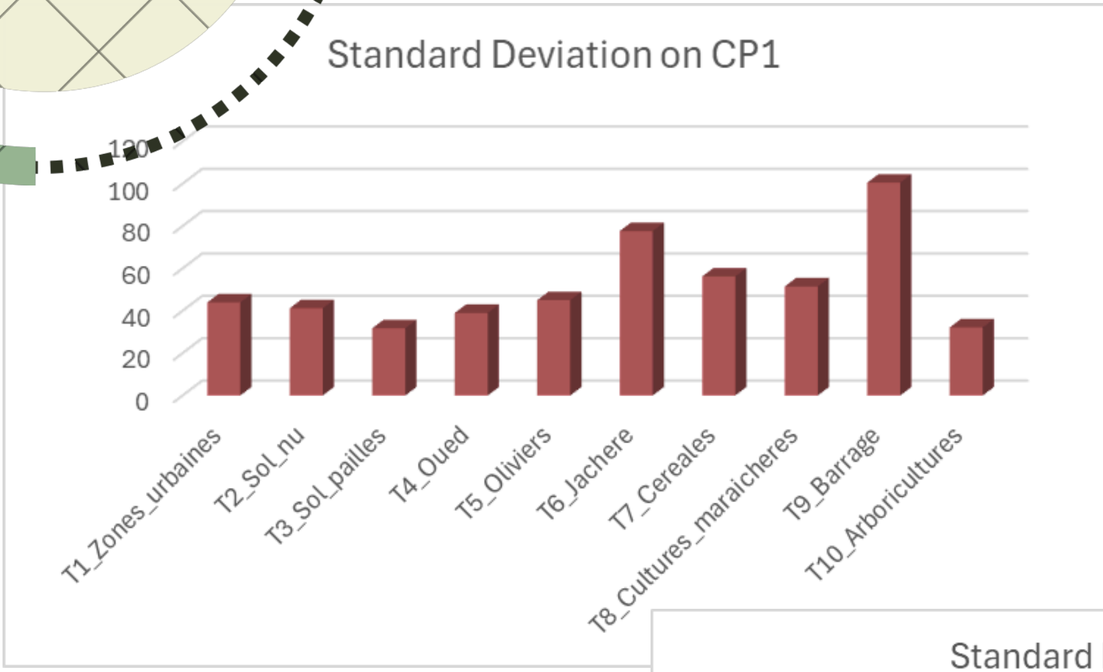


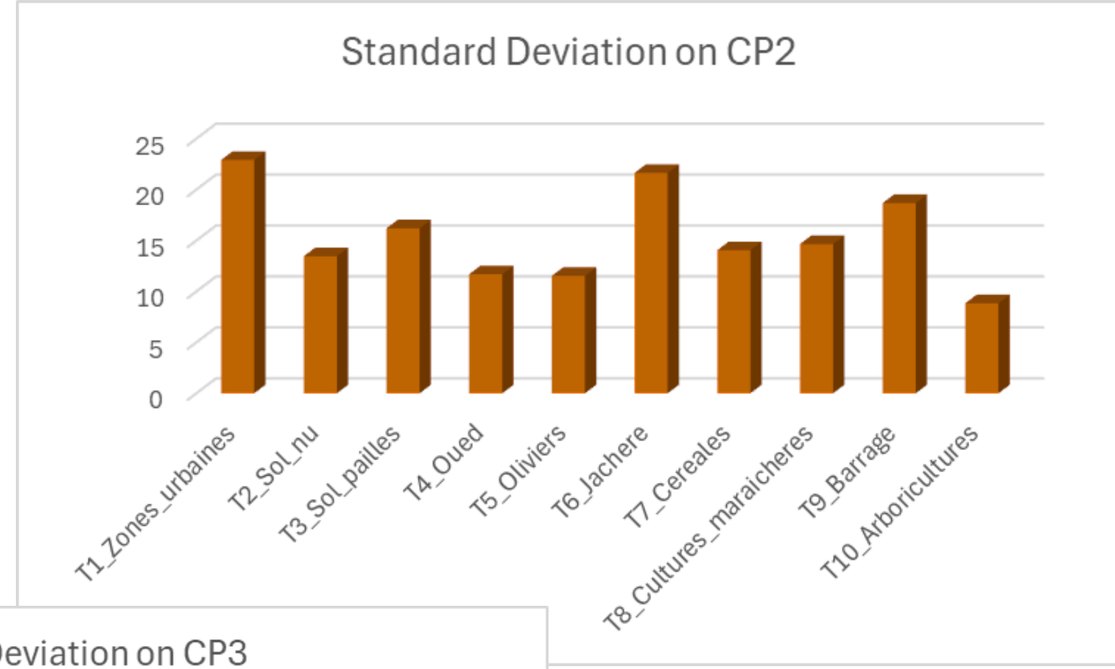


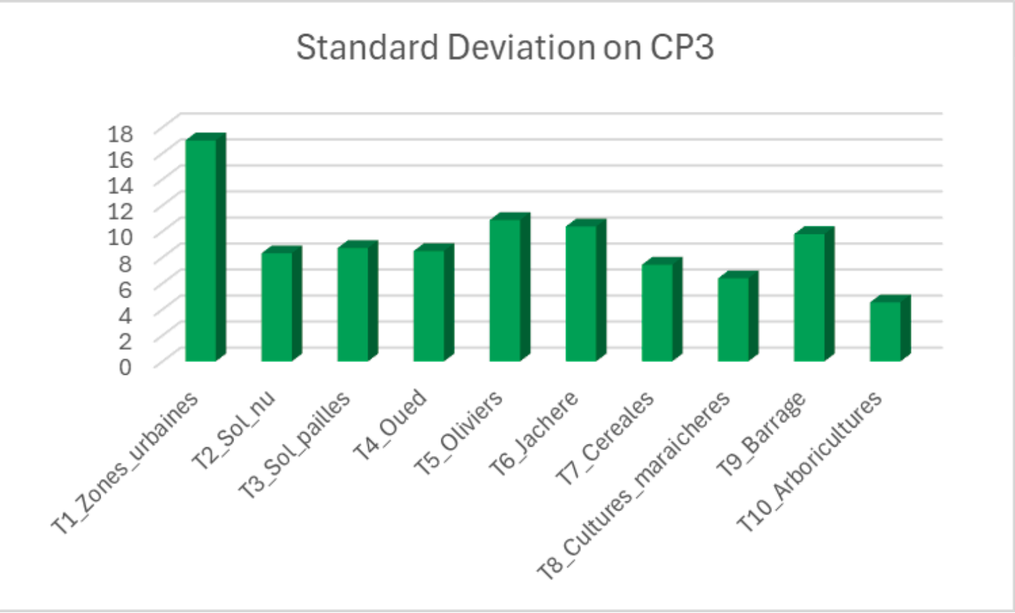


STEP 1.6

Statistical Signatures – PCA Components (CP1, CP2, CP3)

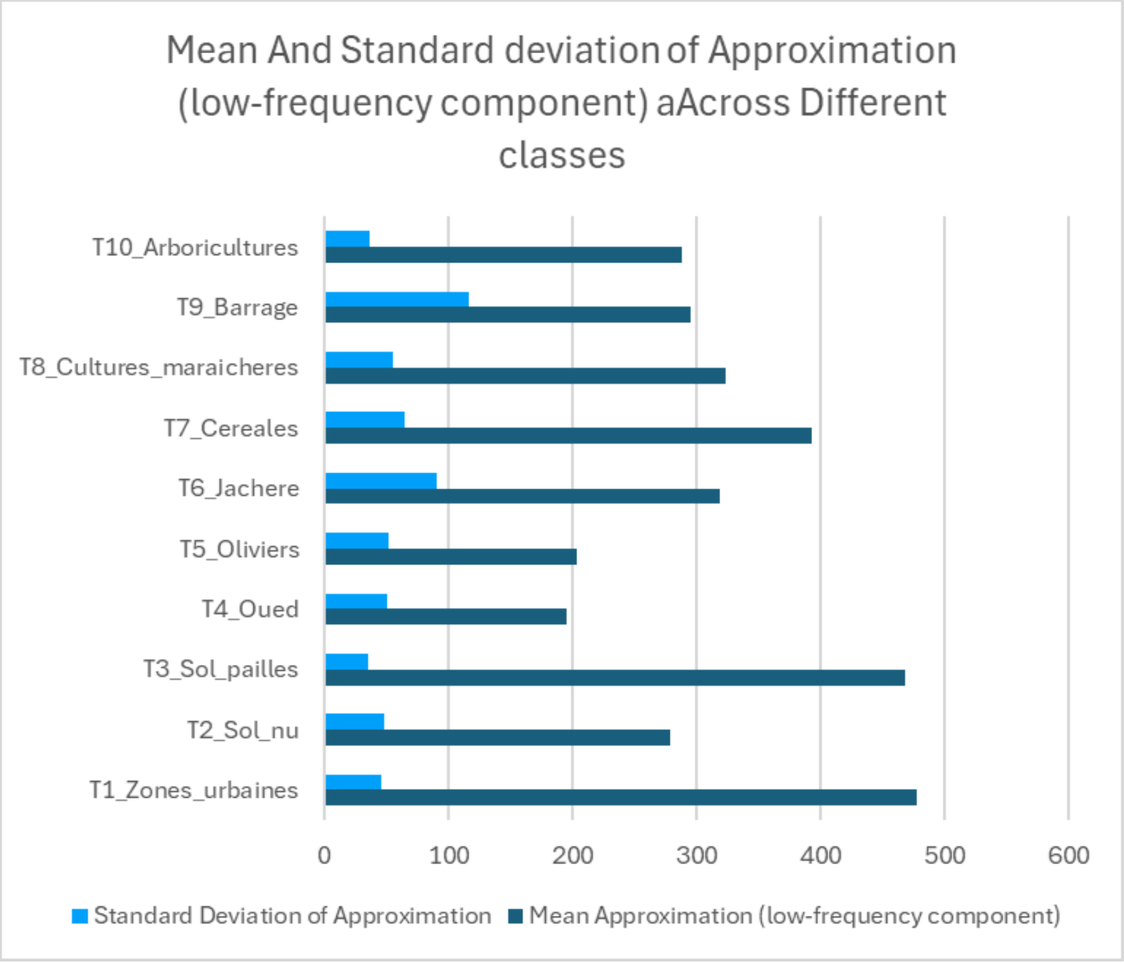


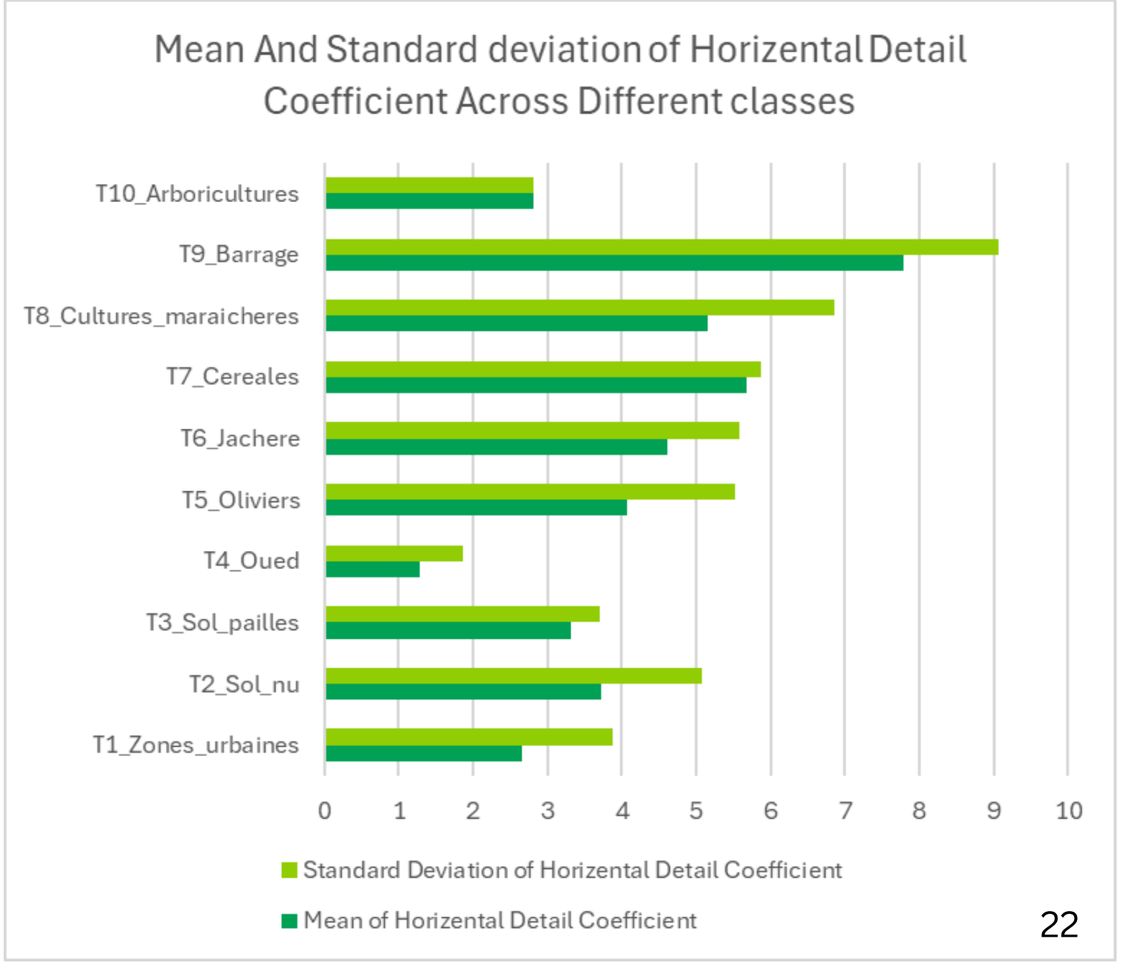




STEP 1.6

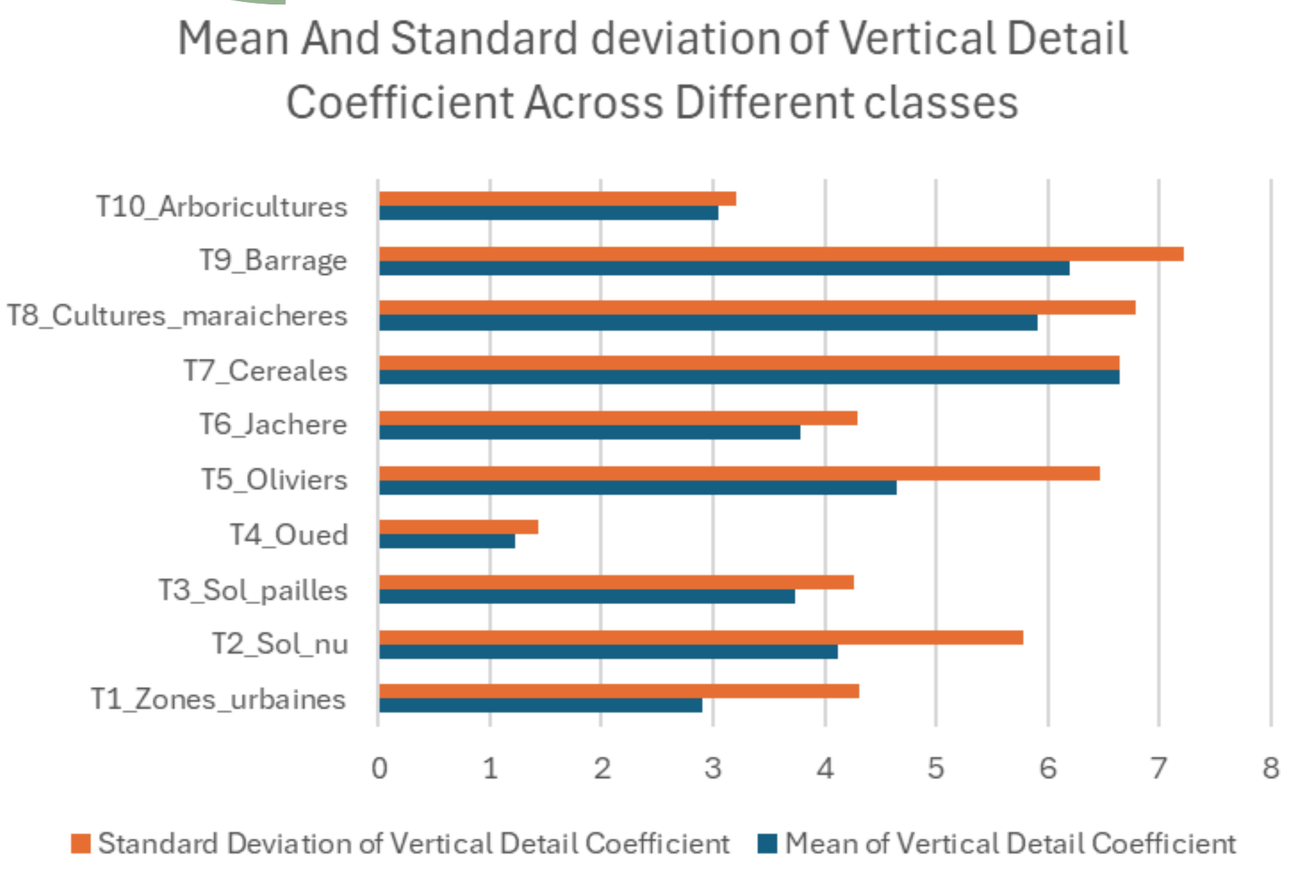
Statistical Signatures – DWT Components (A, DH, DV, DD)

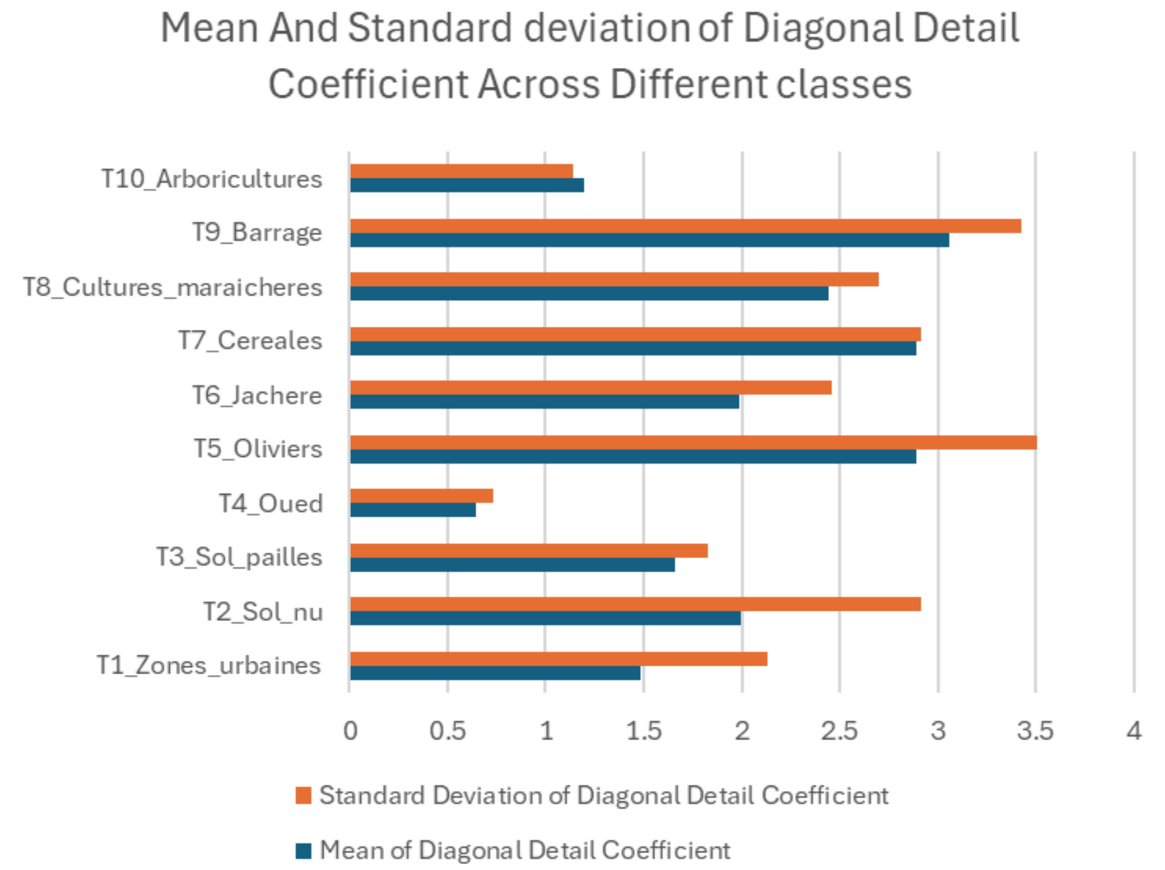




STEP 1.6

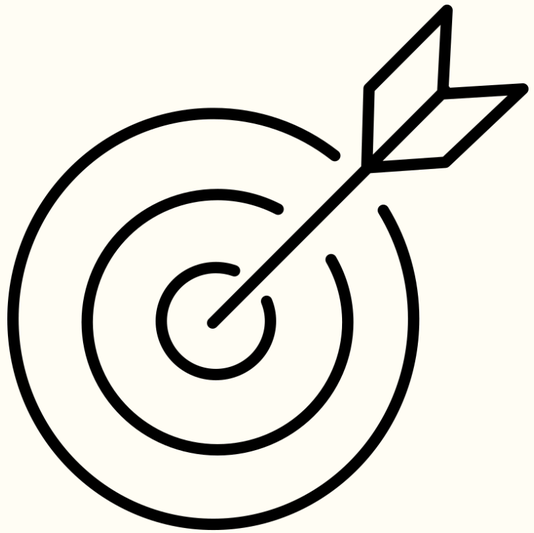
Statistical Signatures – DWT Components (A, DH, DV, DD)

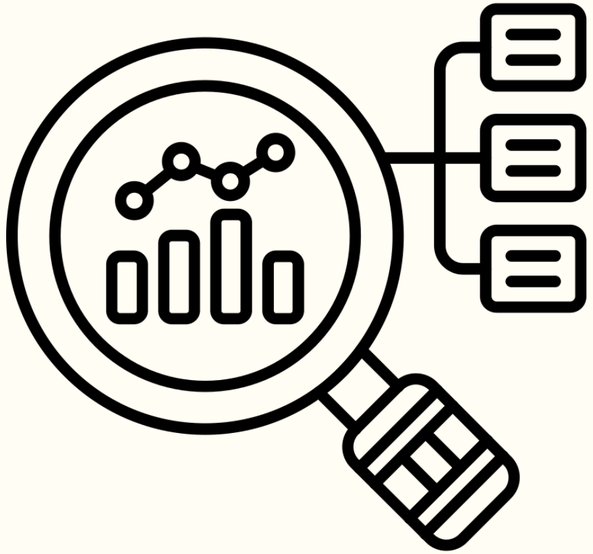




STEP 2

Classification and Accuracy Assessment





STEP 2.1

Functions used in classification

balayagecomplet

Extract local texture features inside moving window

vectcaracteristiqstd4

Builds vertical/horizontal variability vector

tauxclassifT1

Computes confusion matrix & accuracy rate

vectcaracteristiqmoy4

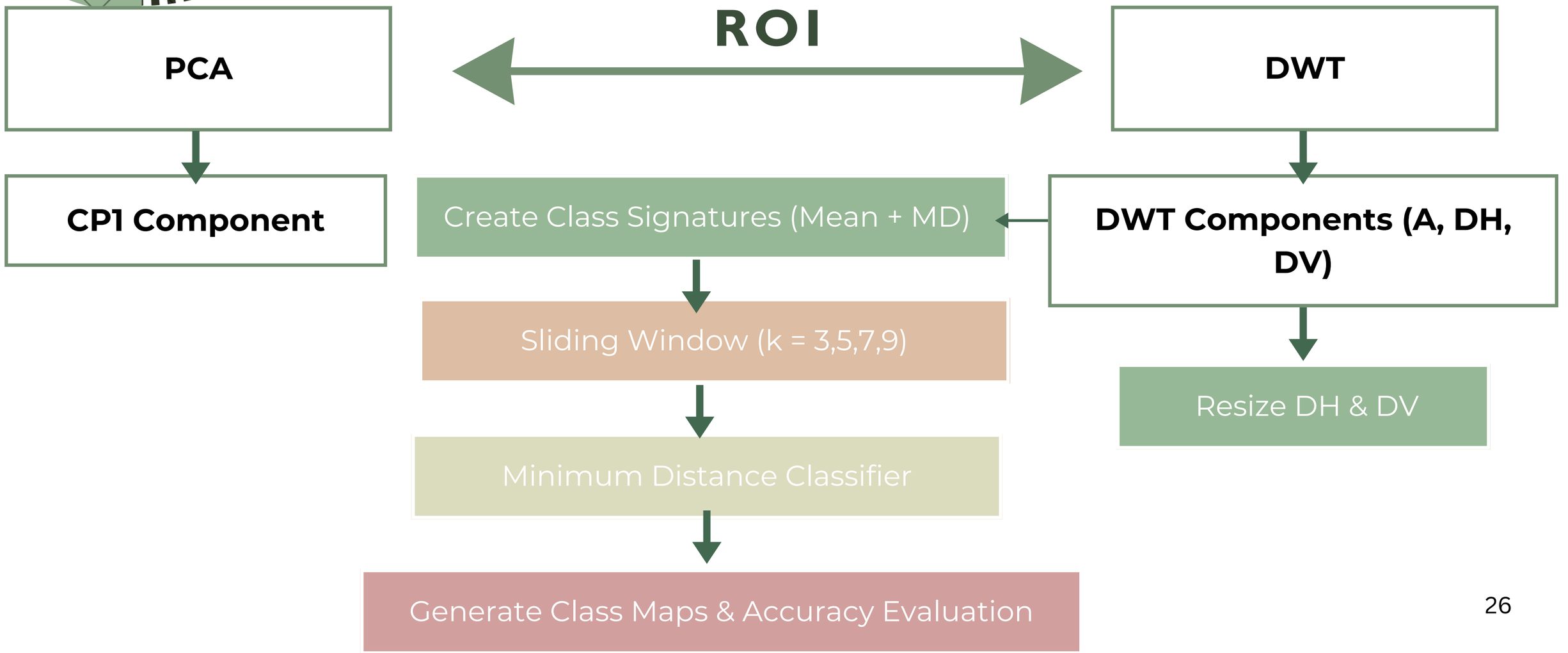
Builds vertical/horizontal mean vector

mindist

Assigns each pixel to closest class based on E, MD

STEP 2.2

Overall Classification Pipeline

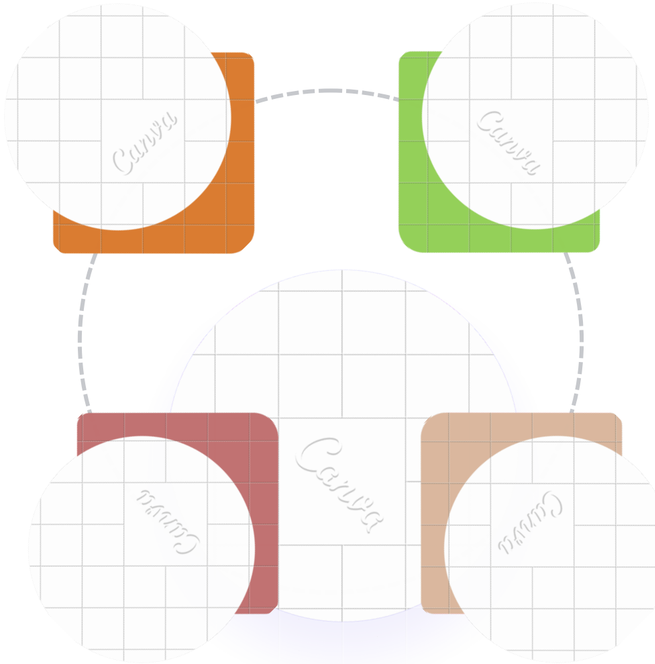


STEP 2.2

Workflow Of PCA Classification

Load CP1 Verify Step 1 files exist CP1 + PCA statistics

Select Sliding Window Size (k = 3,5,7,9)
Extract CP1 window feature
→Minimum Distance to class signature
→ Assign Class Label



Prepare PCA Class Signatures Mean(E_CP1) + MD_CP1 → duplicated → (2×10)signature matrix

Generate PCA Classification Map + Accuracy evaluation

STEP 2.2

Workflow classification of DWT

Load DWT Components (A, DH, DV)

Resize DH & DV to match image size

Verify Step 1 files exist (DWT_results + DWT stats)

Prepare DWT Class Signatures
Mean(DH,DV) + MD(DH,DV) → (2×10) signature matrix

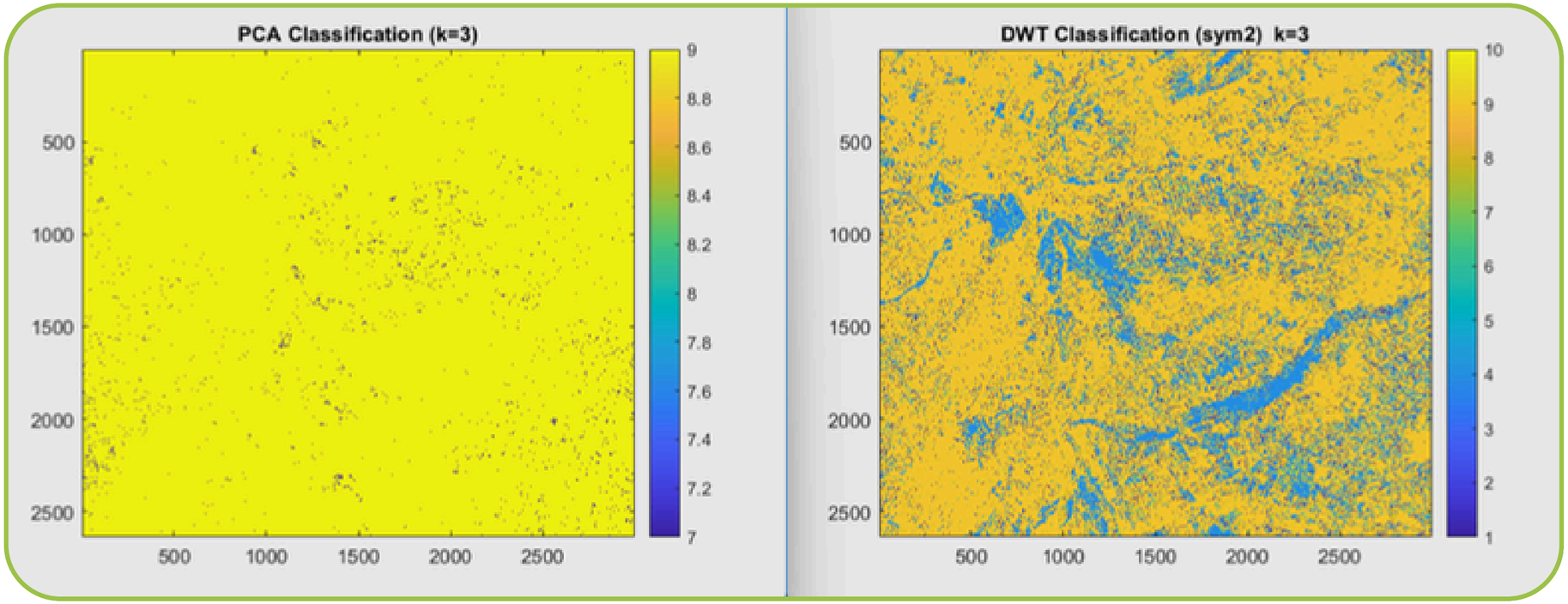
Select Wavelet (sym2 / db45)

FOR each window:
Extract Window Features from DH + DV
→ Minimum Distance to class signature
→ Assign Class Label

Generate DWT Classification Map + Accuracy evaluation

STEP 2.3

PCA Classification Results for K=3

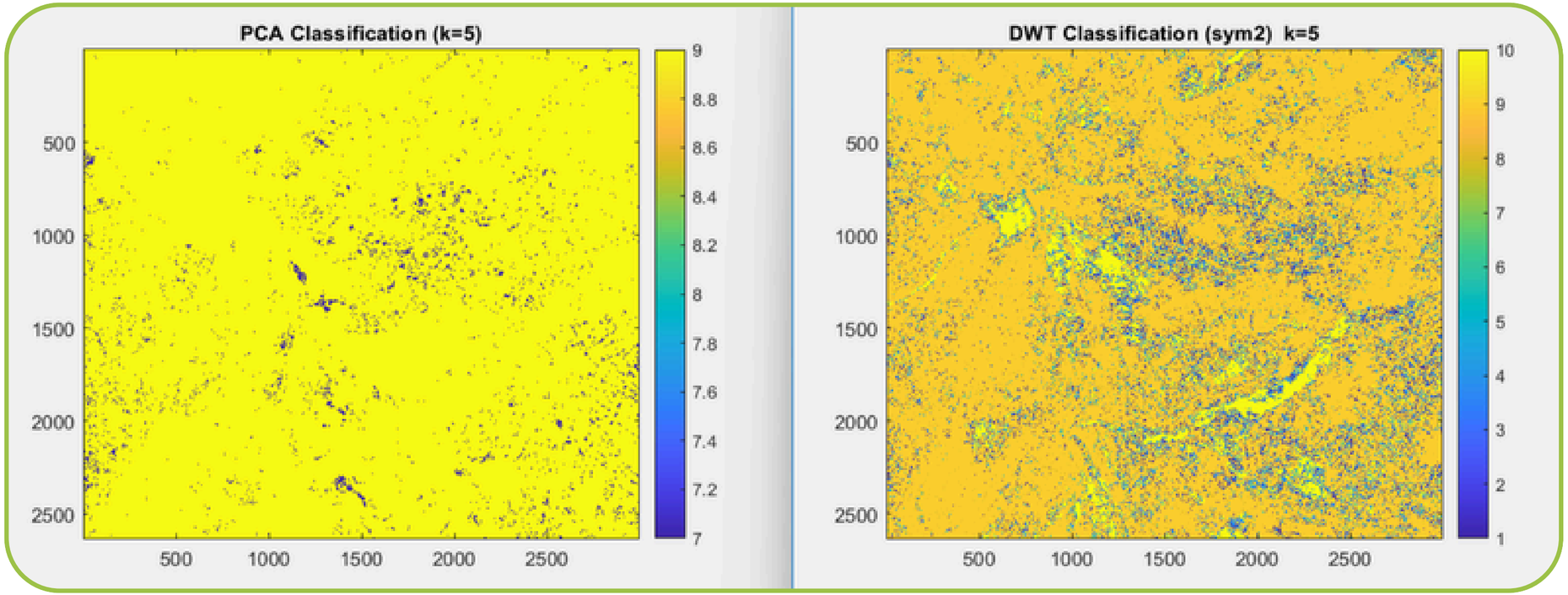


Nearly uniform classification map
No meaningful class separation
Overall Accuracy = 0.00%

```
-- Processing window size k = 3 ---
> PCA (PC1) Classification
> DWT Classification
> Computing confusion matrices
  PCA Overall Accuracy (TCT) : 0.00%
  DWT Overall Accuracy (TCT) : 6.62%
```

STEP 2.2

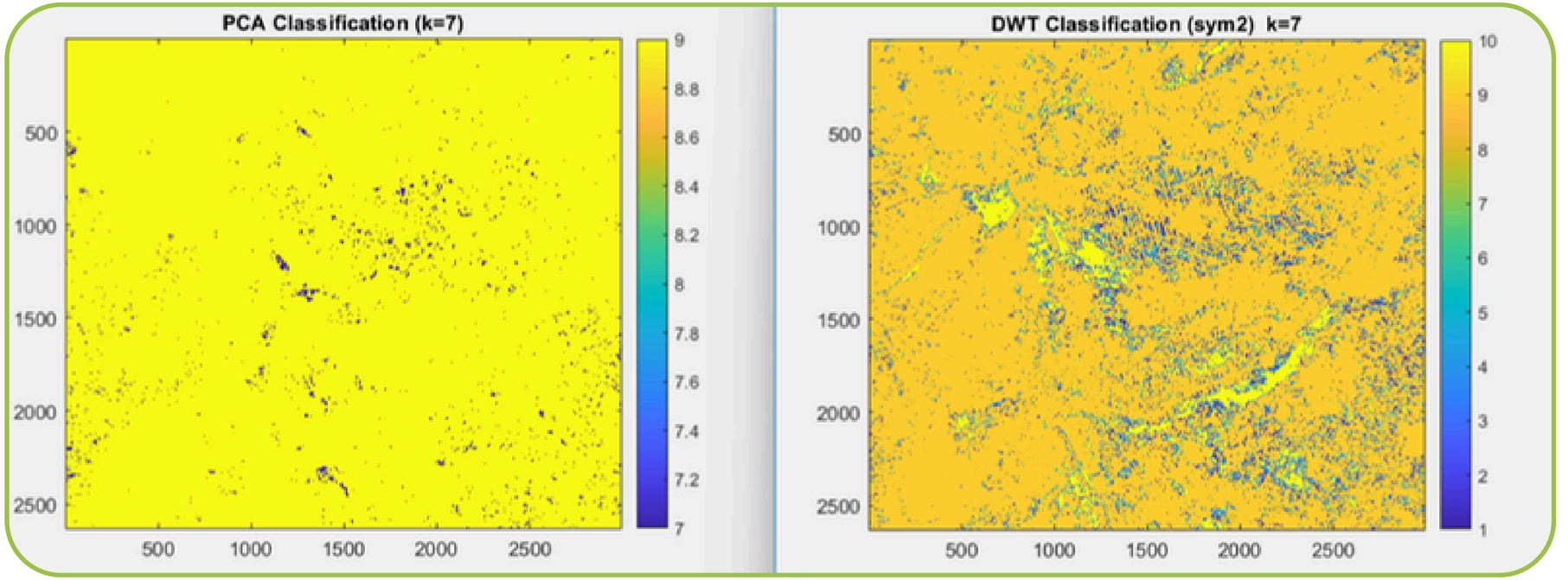
PCA Classification Results for K=5




```
--- Processing window size k = 5 ---
> PCA (PC1) Classification
> DWT Classification
> Computing confusion matrices
  PCA Overall Accuracy (TCT) : 0.00%
  DWT Overall Accuracy (TCT) : 5.24%
```

STEP 2.2

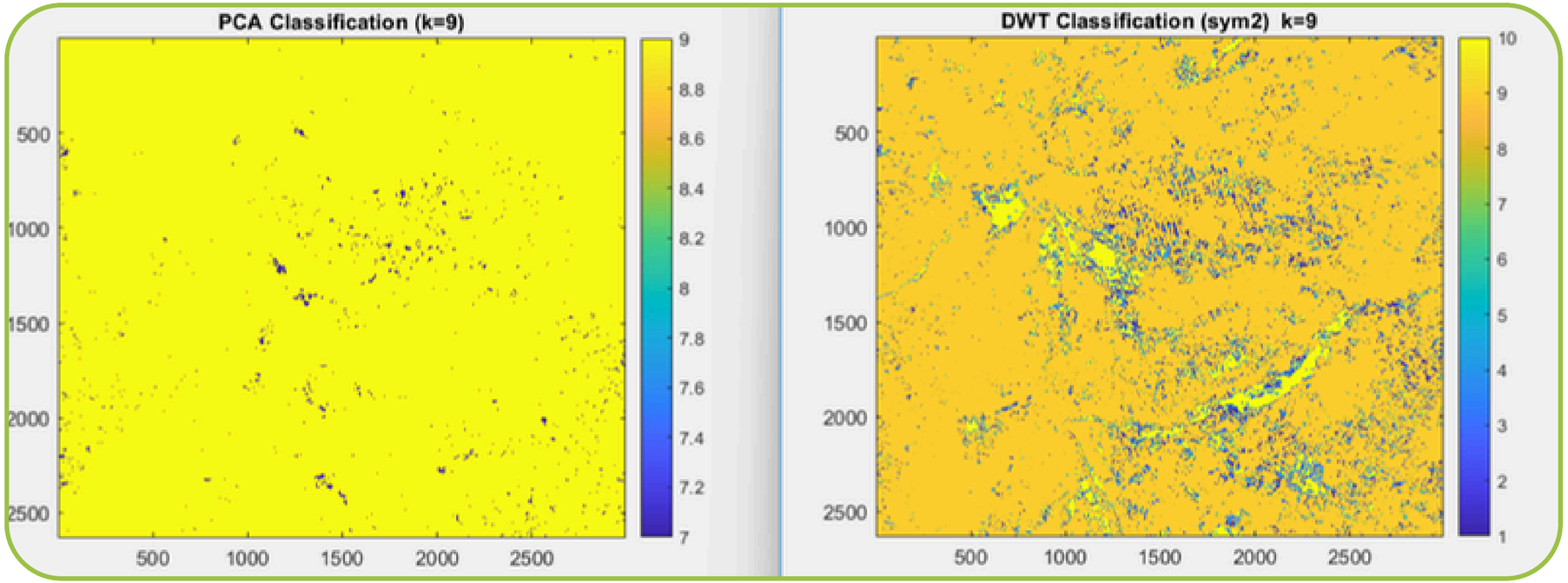
PCA Classification Results for K=7




```
- Processing window size k = 7 ---
PCA (PC1) Classification
DWT Classification
Computing confusion matrices
PCA Overall Accuracy (TCT) : 0.00%
DWT Overall Accuracy (TCT) : 4.30%
```

STEP 2.2

PCA CLASSIFICATION map K=9




```
-- Processing window size k = 9 ---
> PCA (PC1) Classification
> DWT Classification
> Computing confusion matrices
  PCA Overall Accuracy (TCT) : 0.00%
  DWT Overall Accuracy (TCT) : 3.62%
=== STEP 2 COMPLETED SUCCESSFULLY ===
```

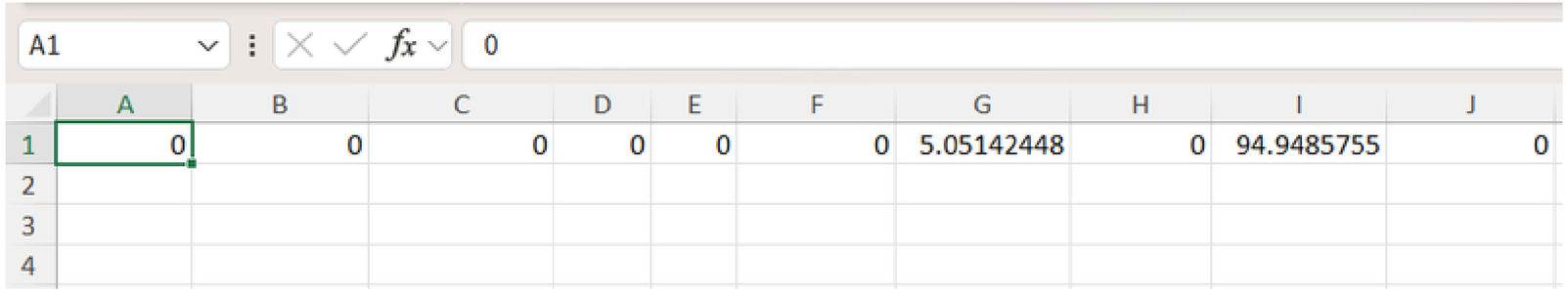
PCA : Nearly entirely uniform yellow - showing virtually no class separation
Overall Accuracy: 0.00% (essentially random/failed classification)
Visual Pattern: The map shows no meaningful spatial structure or class boundaries

DWT:
Unlike PCA, different classes are visibly distinguishable.
Overall Accuracy is better
DWT successfully captures texture and directional information
A small window preserves local details important for classification

STEP 2.2

Comparaison between PCA and DWT Confusion Matrices

PCA for k=3



A1 | 0

	A	B	C	D	E	F	G	H	I	J
1	0	0	0	0	0	0	5.05142448	0	94.9485755	0
2										
3										
4										

DWT for k=3

	A	B	C	D	E	F	G	H	I	J
1	6.61742402	0	2.81318314	6.30627667	3.84226747	0	0	7.75419764	62.7326629	9.93398812
2										

STEP 2.2

Comparaison between PCA and DWT Confusion Matrices

PCA for k=5

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	0	0	0	0	0	0	2.643927634	0	97.35607237	0			
2													
3													
4													
5													

DWT for k=5

	A	B	C	D	E	F	G	H	I	J
1	5.24345082	0	2.1209081	3.9995306	2.13214861	0	0	6.73034092	72.3954363	7.37818462
2										

STEP 2.2

Comparaison between PCA and DWT Confusion Matrices

PCA for k=7

	A	B	C	D	E	F	G	H	I	J	K
1	0	0	0	0	0	0	2.643927634	0	97.35607237	0	
2											
3											
4											
5											

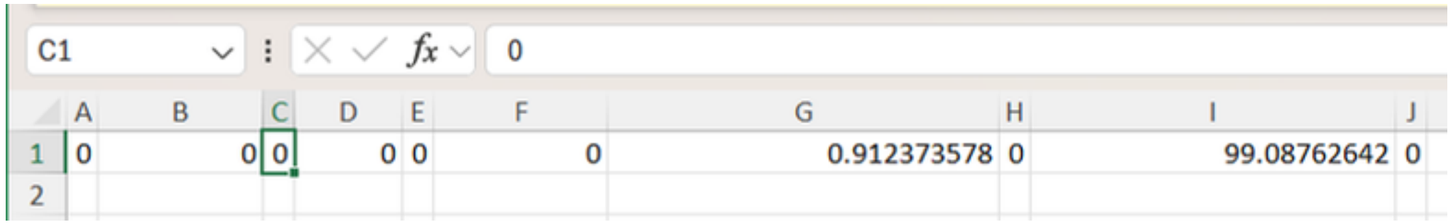
DWT for k=7

	A	B	C	D	E	F	G	H	I	J
1	4.29806678	0	1.71477992	2.93476585	1.41095415	0	0	5.76601443	78.0228037	5.85261512
2										
3										

STEP 2.2

Comparaison between PCA and DWT Confusion Matrices

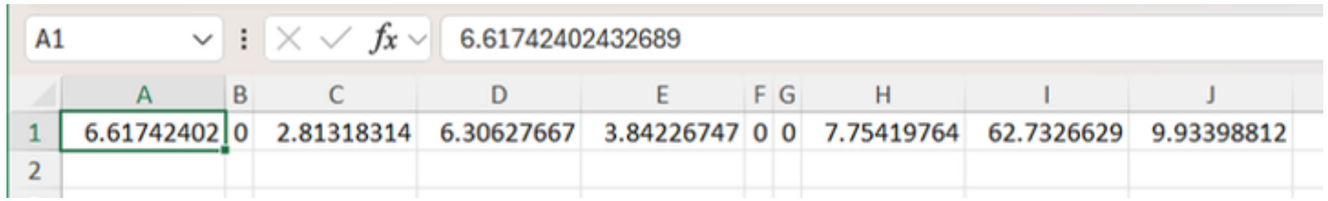
PCA for k=9



C1 | 0

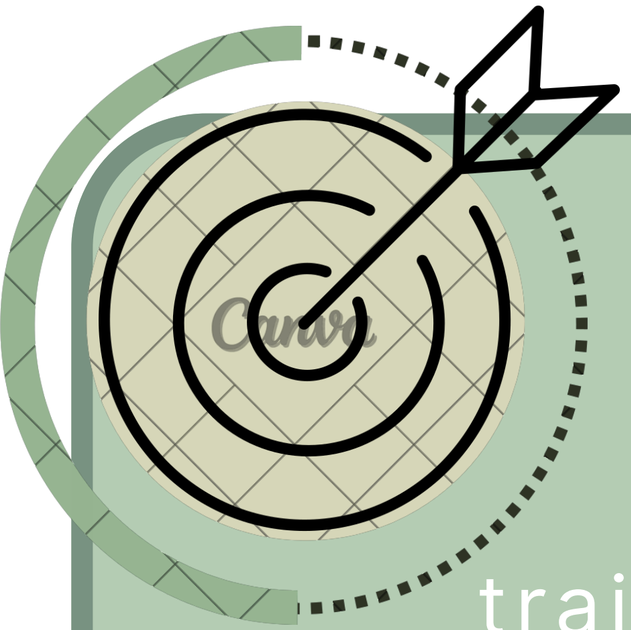
	A	B	C	D	E	F	G	H	I	J
1	0	0	0	0	0	0	0.912373578	0	99.08762642	0
2										

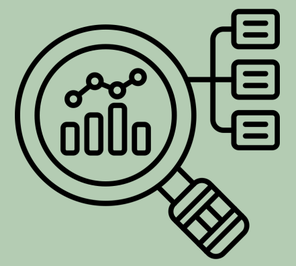
DWT for k=9



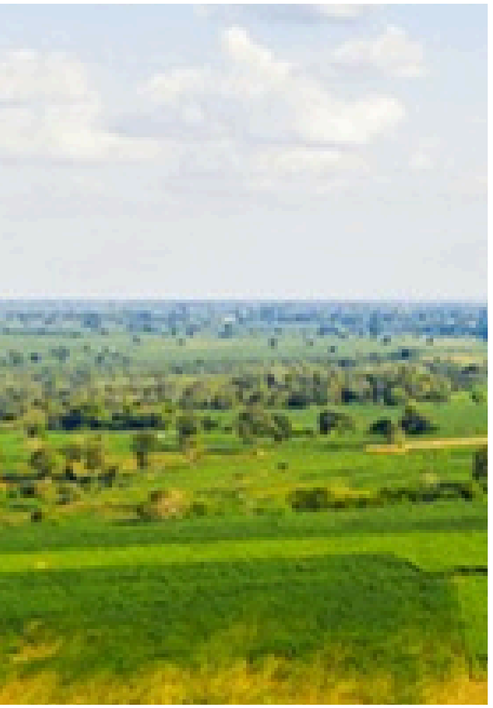
A1 | 6.61742402432689

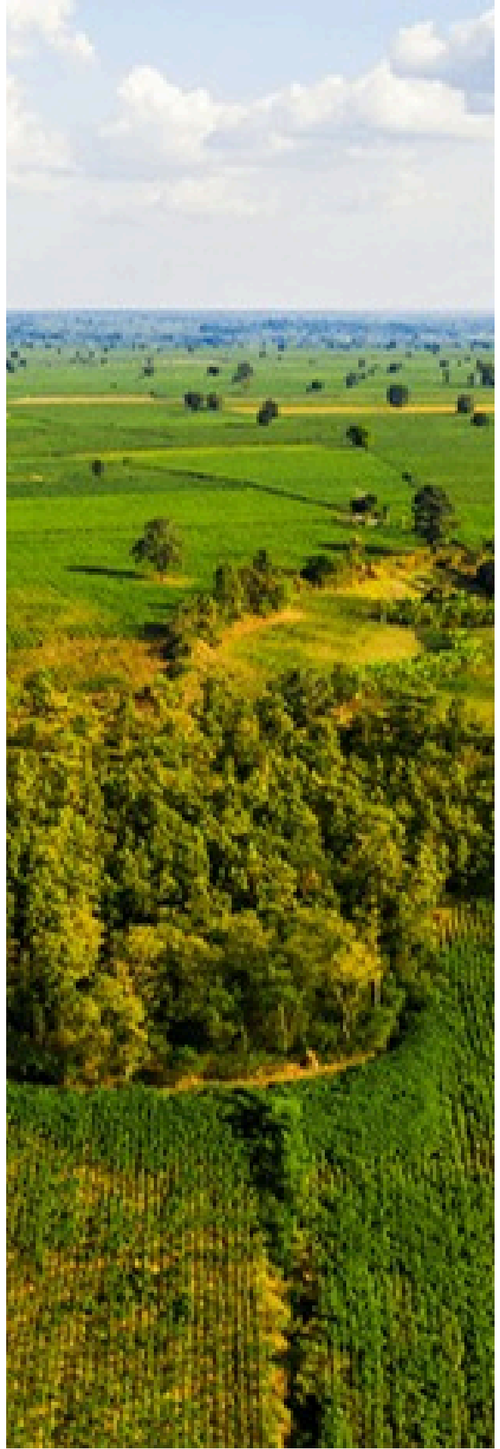
	A	B	C	D	E	F	G	H	I	J
1	6.61742402	0	2.81318314	6.30627667	3.84226747	0	0	7.75419764	62.7326629	9.93398812
2										





Supervised classification requires training data (manual ROIs) to learn class signatures, then applies these to classify the entire image.







CONCLUSION

- This study combined PCA, DWT, and manual ROI-based classification to effectively map land cover in the Kairouan region.
- PCA reduced spectral redundancy, while DWT added texture and spatial detail, improving class separation.
- Manual ROIs ensured accurate training samples, though subjectivity, 10 m resolution, and the use of a single image remain limitations.
- The workflow is reliable and adaptable, offering a solid foundation for land-cover analysis in semi-arid regions.
- Future work could integrate multi-temporal or higher-resolution imagery, additional indices, automated classifiers, and accuracy validation.

Thank You

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